Nathan Riek

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SUMMARY

PhD-trained machine learning engineer and data scientist with 5+ years of experience transforming messy data into production-ready predictive models, with deep experience in physiological time-series data.

EXPERIENCE

Applied ML Scientist | AliveCor | Remote | Jun 2026 – Present

- Designed an ECG beat detection algorithm

Founding Machine Learning Engineer | IntelliCardio | Remote | Sep 2025 – Jun 2026

- Built and deployed an LLM-powered chatbot that answers ECG-related questions using LangChain and RAG, ensuring medically grounded responses with inline source citations (deployed on Streamlit Community Cloud)
- Developed a desktop application for ECG visualization, preprocessing, feature extraction, and ML inference for clinical researchers

Data Scientist | University of Rochester | Rochester, NY | Sep 2025 – Jun 2026

- Deploying an ML clinical decision support tool within a hospital Azure cloud environment
- Developed a survival model (scikit-survival) to predict atrial fibrillation risk up to 1 year before first occurrence
- Adapted ECG preprocessing pipeline from PhD research for compatibility with consumer-grade ECG devices

Data Science Intern | HeartBeam | Remote | Sep 2024 – Dec 2024

- Trained deep learning models (PyTorch) to estimate cardiac age from noisy ECG time-series data
- Automated data preparation and prediction retrieval using Selenium, reducing manual processing time

PhD Researcher | University of Pittsburgh | Pittsburgh, PA | Jan 2021 – Aug 2025

- Designed ECG and EEG preprocessing pipelines using Python and MATLAB, adopted by 5+ research institutions
- Developed ML (PyTorch, Scikit-learn) models for ECG-based classification of occlusion myocardial infarction; recipient of 2024 ISCE Early Career Investigator Award
- Deployed ML decision support platform via Dash and Azure to assist clinicians with ECG interpretation
- Published in top-tier journals (Nature Medicine, European Heart Journal, IEEE TBME)

Engineering Co-Op | Eaton | United States | Sep 2018 – Aug 2020

- Built a C# GUI test automation tool, integrated RF instrumentation into TestStand/LabVIEW workflow, and contributed to Lean Six Sigma initiatives during my three Co-Op rotations

EDUCATION

PhD, Electrical and Computer Engineering
University of Pittsburgh | Jan 2021 – Aug 2025
BS, Electrical Engineering
University of Pittsburgh | Aug 2016 – Dec 2020

CERTIFICATIONS

IBM Data Science Professional Certificate | Coursera | Nov 2025
IBM Generative AI Engineering Professional Certificate | Coursera | Feb 2026

SKILLS

Machine Learning: PyTorch, Scikit-learn, Weights & Biases, MLflow
Programming: Python, SQL, R, MATLAB
Data and Feature Engineering: Pandas, NumPy, SciPy
LLMs & NLP: LangChain, Retrieval-Augmented Generation (RAG), Hugging Face Transformers
Visualization: Plotly, Matplotlib, Seaborn
Dashboards: Dash, Streamlit
Cloud & Deployment: Azure, AWS, Docker
Version Control: Git, GitHub